

Young Innovators · Class 8

RFID Smart Door Lock with Access Log

Tap your card, the door opens. Strangers stay out!

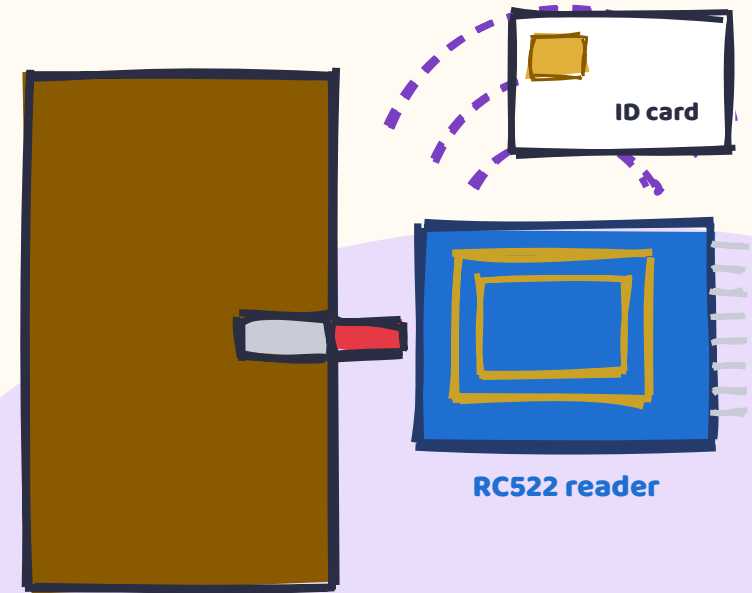
RC522 RFID reader

Allow lists

Privacy & security

Did you know?

Most RFID cards have no battery. The reader's radio field powers the chip for a split second.





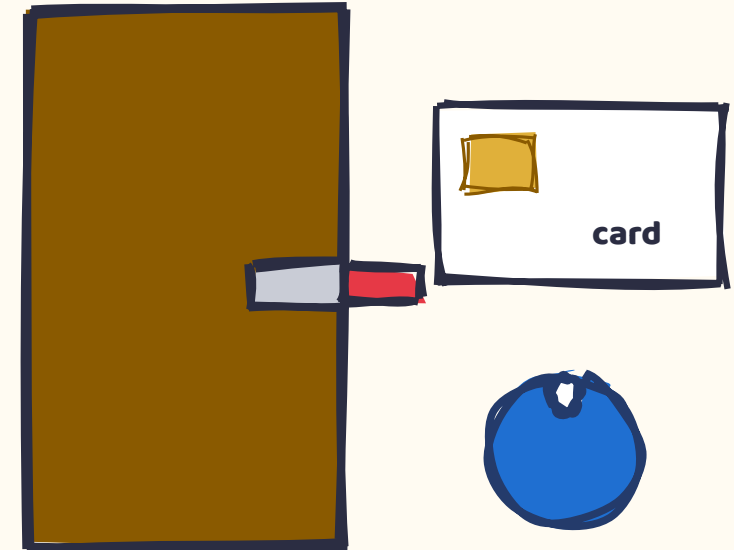
Let's think...

What is wrong with metal keys?



Key problems

Keys get **lost** and copied. If a staff member leaves, you must change **every lock**. And a key can't tell you **who** opened the door, or **when**.



Hide the key?

Change all locks?

Use smart cards!

Today: you build a lock that only opens for **approved cards**, rejects strangers and keeps an **access log**.





Learning aims

Today I will learn to...



1  Explain how **RFID** cards work without batteries

2  Read a card's **UID** with the RC522 reader

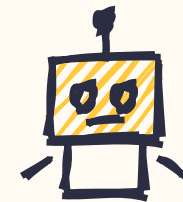
3  Compare the UID with an **allow list** using arrays and loops

4  Unlock a **servo** and show an **access log** on the OLED

5  Discuss **privacy**, lost cards and how secure RFID really is

Keywords

RFID · UID · hexadecimal · array · allow list · log · privacy

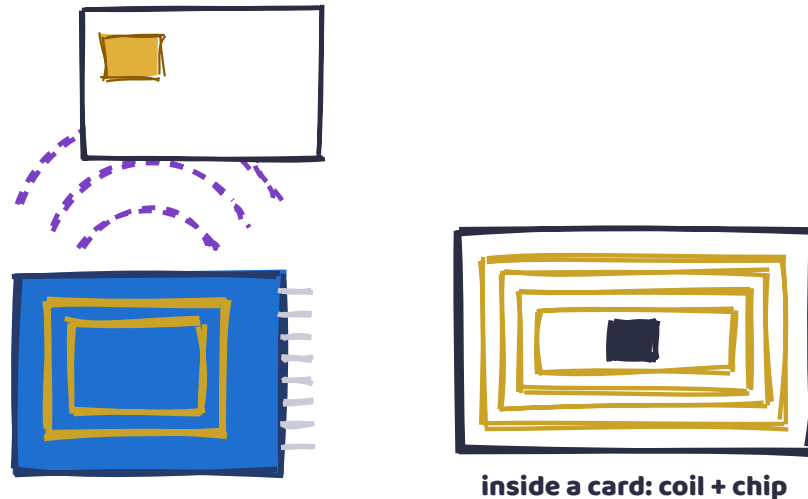




The science



How a card talks without a battery



Step by step

1. The reader makes a **radio field** (13.56 MHz).
2. The card's **coil** catches energy, like a tiny wireless charger.
3. The chip wakes up and sends back its **ID number**.
4. All in a fraction of a second!

Short range on purpose

Our cards only work within a few centimetres. That stops the door opening when you just walk past.

RFID means

Radio Frequency IDentification. NFC on phones is a close cousin.





Real world

Where do we tap cards?



Offices & hotels

Room key cards that can be switched off at reception



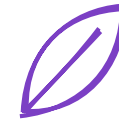
Metro & buses

Tap a travel card at the gate



Contactless pay

Bank cards and phones use NFC



Pets & farms

A tiny tag under the skin identifies lost pets



Which cards or tags do your family use? Do they say WHO you are, or just let you in?

Library books

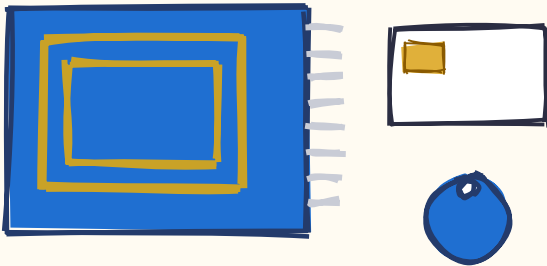
Many libraries put RFID tags in books for quick self-checkout.





Meet the reader

The RC522 RFID module



3.3 V only!

Like the NRF24 radio, the RC522 is powered from **3.3V**. 5V can damage it.

Library

Install **MFRC522** (by GithubCommunity / miguelbalboa) from the Library Manager.

SPI again

It uses the same **SPI** pins (11, 12, 13) as the radio in YI-10. **SDA** here means chip-select, not I2C!

Cards + fobs

The kit comes with a white card and a blue key fob. Both work the same way.

3.3V	3.3V (NOT 5V!)
RST	pin 9
GND	GND
MISO / MOSI	12 / 11
SCK	13
SDA (SS)	pin 10





Big idea

Every card has an ID called a UID

**A3 4F 1B 9C**4 bytes, written in **hexadecimal**

Unique-ish

Cheap cards have 4-byte UIDs: about 4 billion possible numbers. Collisions are rare, but copies are possible (see slide 16!).

Hex digit	Normal number
0 to 9	0 to 9
A	10
B, C, D, E	11, 12, 13, 14
F	15
FF	255 (biggest byte)

Why hex?

One byte (0 to 255) always fits in exactly **2 hex digits**.
Easy to read and compare!

Try it

What is 2F in normal numbers? ($2 \times 16 + 15$)





Class game

The bouncer



How to play

1. One student is the **bouncer** holding the secret **allow list** of 4 ID numbers.
2. Everyone else gets a card with a number.
3. Show your card. On the list? Come in, and the bouncer writes your name and time in the **log**.
4. Not on the list? Rejected!
5. Then: someone copies a valid number onto a blank card...

Then discuss

Could the bouncer tell the copied card from the real one? What does that mean for our lock?



Big idea

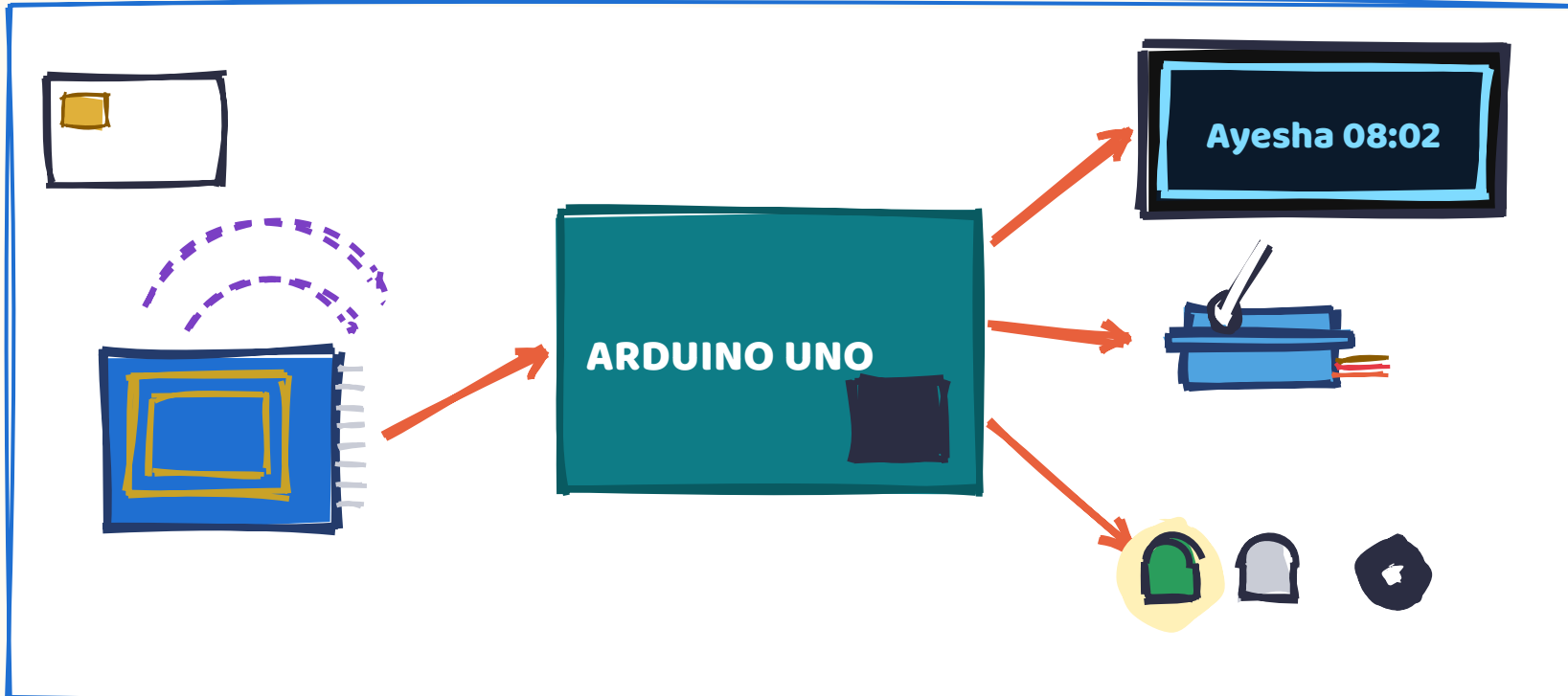
Checking an ID number is only as safe as the number is hard to copy.





The plan

Read, check, decide, log



Two outcomes

Approved: green LED, servo unlocks for 3 s, name in log.

Unknown: red LED, 2 low beeps, UID logged as 'unknown'.

Servo = bolt

0° locked, 90° open. Fix a card 'bolt' to the horn.



Build it



Our smart lock kit (per group)



Arduino Uno

The brain



RC522 + card + fob

The reader



Servo SG90

The bolt



OLED 0.96 in

Access log



Green + red LED

With $2 \times 220 \Omega$



Buzzer

Reject sound



Also: breadboard, M-F jumpers for the RC522, a card box door for the model, laptop with Arduino IDE 2.

Bring cards!

Old hotel keys or metro cards may also work, if they use 13.56 MHz.



Build it

Wiring the smart lock



Part	Uno
RC522 3.3V / GND	3.3V / GND
RC522 SDA / RST	10 / 9
RC522 SCK / MOSI / MISO	13 / 11 / 12
Servo signal (orange)	pin 3
Servo red / brown	5V / GND
Green / red LED (+ 220 Ω)	pin 4 / pin 5
Buzzer +	pin 8
OLED SDA / SCL	A4 / A5



Check before USB

RC522 on **3.3V**? Servo on **5V**? Swapping them is the most common mistake.



Servo jitter?

If the Uno resets when the servo moves, power the servo from a separate 4×AA pack (share GND!).



Code it: part 1

Find your card's UID



```
#include <SPI.h>
#include <MFRC522.h>
MFRC522 rfid(10, 9);          // SS (SDA), RST

void setup() {
  Serial.begin(9600);
  SPI.begin();
  rfid.PCD_Init();
  Serial.println("Tap a card...");
}

String readUID() {
  String id = "";
  for (byte i = 0; i < rfid.uid.size; i++) {
    if (rfid.uid.uidByte[i] < 0x10) id += "0";
    id += String(rfid.uid.uidByte[i], HEX);
    if (i < rfid.uid.size - 1) id += " ";
  }
  id.toUpperCase();
  return id;
}

void loop() {
  if (!rfid.PICC_IsNewCardPresent()) return;
  if (!rfid.PICC_ReadCardSerial()) return;
  Serial.println(readUID());
  rfid.PICC_HaltA();          // stop reading this card
}
```

Do this first!

Upload, open the **Serial Monitor**, tap each card and fob.
Write down every UID. You need them for the allow list.

Why add a 0?

Byte 0x0A would print as just **A**. Adding the 0 makes it **0A**, so every UID has the same format.





Code it: part 2

The allow list and the bolt



```
#include <Servo.h>
Servo bolt;
String allowed[] = {"A3 4F 1B 9C", "04 7A 22 B1"}; // YOUR UUIDs
String names[] = {"Ayesha", "Bilal"};
const int COUNT = 2;

// in setup(): bolt.attach(3); bolt.write(0);

// in loop(), after reading the UID:
String id = readUID();
int who = -1; // -1 = not found
for (int i = 0; i < COUNT; i++) {
  if (id == allowed[i]) who = i;
}
if (who >= 0) {
  digitalWrite(4, HIGH); // green
  bolt.write(90); // unlock
  delay(3000);
  bolt.write(0); // lock again
  digitalWrite(4, LOW);
} else {
  digitalWrite(5, HIGH); // red
  tone(8, 200, 300); delay(400); tone(8, 200, 300);
  delay(1000);
  digitalWrite(5, LOW);
}
```

Two arrays, same order

allowed[0] belongs to **names[0]**. The loop finds the position **i** of the matching UID, so we know the name too.

Why -1?

Positions start at 0, so -1 can never be a real position. It means 'not found'.





Code it: part 3

An access log on the OLED



```
String logNames[3] = {"-", "-", "-"};
unsigned long logTime[3] = {0, 0, 0};

void addLog(String who) {
  for (int i = 2; i > 0; i--) {           // shift old entries down
    logNames[i] = logNames[i - 1];
    logTime[i] = logTime[i - 1];
  }
  logNames[0] = who;
  logTime[0] = millis() / 60000;          // minutes since power on
  showLog();                             // draw 3 lines on the OLED
}
```

Newest on top

Each new entry pushes the others **down**. The oldest falls off the end. This is called a **rolling log**.

No clock?

The Uno does not know the real time. We show **minutes since power on**. A real-time clock module would fix this.

Unknown cards too

Call `addLog("UNKNOWN")` for rejected cards. Security staff want to know!

ACCESS LOG

Ayesha 12 min

UNKNOWN 9 min

Bilal 3 min



Test like an engineer

Does the lock work?

Test	Expect	Result
Approved card	green, unlock 3 s, name logged	pass / fail
Approved fob	same as above	pass / fail
Unknown card	red, 2 beeps, UNKNOWN logged	pass / fail
Hold a card there for 5 s	only ONE unlock	pass / fail
Card 6 cm away	nothing	pass / fail

Debug checklist

No UID shown? Check 3.3V, SDA on 10 and RST on 9.

Approved card rejected? Copy the UID exactly, spaces and capitals included.



Print the UID AND the comparison result to the Serial Monitor while testing.



Think like a security engineer

How safe is our lock really?



UIDs can be copied: cheap 'magic' cards can clone a UID. Banks and offices use encrypted cards instead.



Lost card: remove its UID from the list straight away. No need to change the lock!



The log is personal data: it shows where people were and when. Who should be allowed to read it?



Fail safe: in a fire, should the door unlock automatically? Real buildings say yes.



Be fair: tell people that entries are logged, and delete old logs.



Discuss

Should a school log every time a student enters the library? Pros and cons?



Level up!

Smart lock challenges

Choose your challenge



Master card: tapping the master card, then a new card, adds the new card to the list.



Lockout: 3 unknown cards in a minute = ignore all cards for 30 seconds and sound an alarm.



Remember: save the allow list in EEPROM so new cards survive a power cut.



Time rules: some cards only work during school hours (needs a real-time clock module).

Think like an engineer

Our list lives inside the lock. How do big buildings with 500 doors update every lock when someone leaves?

Boss level

Locks connect to a central server over the network. One change updates every door!



Show what you know

What did we learn?

- ☒ RFID cards get power from the reader's radio field
- ☒ Each card has a UID, shown in hexadecimal
- ☒ An allow list + loop decides who may enter
- ☒ A rolling log records who entered and when
- ☒ UID-only locks can be cloned, and logs are personal data

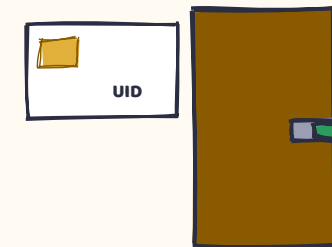


Exit ticket: why do we use -1 to mean 'not found'?



At home

Count how many RFID or NFC cards your family uses. What does each one open or pay for?



Next class: a
Rechargeable
Power Pack for
Robots!

